

Teaching Statement

“Education is not preparation for life; education is life itself.” — John Dewey

I believe education can transform lives in ways that may not be visible immediately but become clear over time. Even a single lecture or discussion can plant a seed that grows into a lifelong pursuit. This perspective, echoed by John Dewey, inspires my teaching philosophy: I treat every session as an opportunity for lasting impact. I relate to this through my own undergraduate experience, when professors in Human–Computer Interaction (HCI) shaped my enthusiasm for the field. Their passion—far beyond the slides—left a lasting impression that continues to guide how I want to teach.

As part of my teaching and assessment repertoire, I hope to introduce and further develop the Cambridge tutorial system of **supervisions**¹, particularly for small-group courses. This student-led and more democratic form of mentoring allows 1-3 students to request meetings on topics they wish to explore in greater depth. The supervisor serves as a guide, and together they agree on an assignment that is reviewed in the next supervision. This structure provides immediate feedback, encourages students to defend and refine their ideas, and can be adapted to the local context to enhance inclusivity and learning outcomes.

At Trinity College, Cambridge, I have solely led and supervised more than eight groups of undergraduates in the Department of Computer Science and Technology for (1) *Further Human–Computer Interaction* (Principal Lecturer: Professor Alan Blackwell) and (2) *Interaction Design* (Principal Lecturer: Professor Hatice Gunes). In addition, I have served as a day-to-day supervisor for Final Year Projects (known as “Part II Projects” at Cambridge) with Professor Cengiz Öztireli.

Beyond Cambridge, my global research collaborations have enabled me to advise more than ten students on their research projects, including some who were just beginning their research journey. While I am proud of these publication outcomes, I see them primarily as evidence of students’ growth and confidence in research, rather than as ends in themselves. Within six months, **all** of them successfully produced publishable work and gained the confidence to present it (e.g., [8, 10, 9]) or at least one co-authored paper/poster with me at top venues (e.g., [7, 1, 6, 4, 2, 3]). Many of these students have since embarked on research careers worldwide with full scholarships. I continue to collaborate with them today, and I take great pride in supporting their growth as independent researchers.

TEACHING SCOPE

I am prepared to independently teach courses such as *Introduction to Virtual and Mixed Reality* and *Designing Interaction Techniques*. I am also open to teaching more foundational courses, for example, *Introduction to Unity Programming*, as well as advanced electives such as *2D/3D Game Design and Interactions*. Across these offerings, I emphasize project-based learning, continuous assessment, and dissemination at academic conferences and public festivals. For example, ACM CHI PLAY hosts an annual Student Game Design Competition, in which I have been a finalist twice [5, 11]. Many other ACM and IEEE conferences also feature student competitions in research, design, and engineering. Success at these venues not only builds students’ confidence but also provides international recognition that can serve as a valuable credential for their future careers.

¹University of Cambridge—What is supervision?

In addition, I can contribute collaboratively to courses such as *Introduction to Human-Computer Interaction* (with other HCI faculty), *Designing Intelligent Interactive Systems* (with AI/HCI faculty specializing in Human-AI interaction), and *Ubiquitous Computing* (with AI/Robotics/HCI faculty working on wearables and ubiquitous computing).

DIVERSITY & INCLUSION

I am committed to creating an inclusive and supportive learning environment where students from diverse backgrounds feel encouraged to participate and succeed. In my teaching, I aim to recognize and value the different experiences, identities, and perspectives that students bring to the classroom. I design activities that allow multiple modes of engagement so that students with varied strengths can contribute meaningfully.

REFLECTIONS

For me, understanding and curiosity matter more than output counts. In supervision, I guide students to pursue work they can be proud of rather than chasing numbers. A conversation with a professor from the University of Tokyo crystallized this philosophy: I evaluate research by three criteria: (1) whether I value it myself, (2) whether I can convince others of its worth, and (3), as a bonus, whether it achieves real-world impact. The same principles shape my teaching: I want students to pursue meaningful work, learn to defend its value, and, when possible, aim for broader impact.

References

- [1] **Li, Xiang**, Chen, Yuzheng, and Tang, Xiaohang. Gesplayer: using augmented gestures to empower video players. In *Companion proceedings of the 2022 conference on interactive surfaces and spaces*, pages 4–8, 2022.
- [2] **Li, Xiang**, Chen, Yuzheng, and Tang, Xiaohang. Gesmessages: Using mid-air gestures to manage notifications. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction*, pages 1–2, 2023.
- [3] **Li, Xiang**, He, Wei, and P. O. Kristensson. Evaluating the usability of microgestures for text editing tasks in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 2026.
- [4] **Li, Xiang**, He, Wei, Jin, Shan, J. Gugenheimer, P. Hui, H.-N. Liang, and P. O. Kristensson. Investigating creation perspectives and icon placement preferences for on-body menus in virtual reality. *Proceedings of the ACM on Human-Computer Interaction*, 8(ISS):236–254, 2024.
- [5] **Li, Xiang**, Tang, Xiaohang, X. Tong, R. Patibanda, F. F. Mueller, and H.-N. Liang. Myopic bike and say hi: games for empathizing with the myopic. In *Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play*, pages 333–338, 2021.
- [6] **Li, Xiang**, Wang, Jin-Du, J. J. Dudley, and P. O. Kristensson. Swarm manipulation in virtual reality. In *Proceedings of the 2023 ACM symposium on spatial user interaction*, pages 1–11, 2023.
- [7] **Li, Xiang**, Wang, Jin-Du, J. J. Dudley, and P. O. Kristensson. Swarm manipulation: An efficient and accurate technique for multi-object manipulation in virtual reality. *Computers & Graphics*, 125:104113, 2024.
- [8] He, Wei, **Li, Xiang**, S. Xu, Y. Chen, C.-I. Sio, G. L. Kan, and L.-H. Lee. Metadragonboat: Exploring paddling techniques of virtual dragon boating in a metaverse campus. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 6335–6344, 2024.
- [9] Wang, Hanbing, J. Du, Y. Li, L. Zhang, and **Li, Xiang**. Grand challenges in immersive technologies for cultural heritage. *International Journal of Human-Computer Interaction*, pages 1–22, 2025.
- [10] Wang, Jin-Du, K. Zhou, H. Ren, P. O. Kristensson, and **Li, Xiang**. Handows: A palm-based interactive multi-window management system in virtual reality. *IEEE Transactions on Visualization and Computer Graphics*, 2025. To appear.
- [11] W. Xu, H.-N. Liang, X. Ma, and **Li, Xiang**. Virusboxing: A hiit-based vr boxing game. In *Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play*, pages 98–102, 2020.